

Coming soon: robots that can touch and feel

Indian-origin researcher Dr Ravinder Dahiya, a professor of electronic and nanoscale engineering, and Professor Duncan Gregory, chair in Inorganic Materials in the School of Chemistry at the University of Glasgow have now ventured on quite an adventure: to make robots go all touchy-feely, that is! They are currently working on a technology to create an ultra-flexible tactile skin for robots that could bring them a step closer to being more human!

The researchers are looking to combine nanotechnology with robotics and develop a printing technique for high-mobility materials such as silicon. Once successful, they will then embed electronics and sensors on a bendable silicon-based surface approximately 50 micrometres thick, or even less. The researchers plan to cover the robot's entire body, so it can feel and touch throughout rather than just restricting to its hands.



Robots that can touch and feel

Likewise, Engineered Arts Limited has come up with a new robot that features a dynamic face reflecting varied moods and even responding accurately to expressions of individuals giving it a look. SociBot-Mini is based on a fully-integrated LED pico-projection technology featuring custom-developed optics.

SociBot-Mini can find and track faces, and also accurately tell an individual's gender and estimate his/her age. These are achieved through an image processing software called SHORE. The robot makes eye contact without having to be prompted and attends to users' needs by analysing their body language. The SociBot features an HD RGB camera, a fully-articulated neck, an IR depth sensor, and sports high-quality audio and a public-facing touchscreen interface.

Nickel-fluoride film to power flexible electronics

Researchers have taken the first step towards creating bendable and fully-foldable electronic devices. In a bid to replace the traditional polymers and carbon-based materials, such as carbon nanotubes, deployed currently to provide critical back-up power for portable and flexible electronics, researchers at the University of Houston have now developed a new 'thin' film for energy storage. The film is much thinner than paper, is flexible as well as bendable and can store enough energy to power bendable and fully foldable electronic devices in the near future.

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